Stochastic Calculus for Quants

Intuition, Memory Tricks, and Desk-Level Logic
From Brownian Motion to Pricing

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Module 1: Why Stochastic Calculus Exists—The Failure of Determinism

1.1 Deterministic Calculus Dies in Markets

Deterministic Model:

Core Formula

$$\frac{dS}{dt} = \mu S \quad \Longrightarrow \quad S(t) = S_0 e^{\mu t}$$

"Deterministic Models Assume Prophets Exist" **Memory:** If markets followed this, we'd all be billionaires. It assumes perfect foresight—no noise, no uncertainty. **Reality:** Prices are continuous but *infinitely rough*.

Why It Fails:

1. **No uncertainty:** Predicts exact path, not distribution
2. **Constant growth:** Ignores volatility clustering
3. **Smooth paths:** Real prices are nowhere differentiable

Market Reality:

$$\Delta S = \mu S \Delta t + \sigma S \epsilon_t \sqrt{\Delta t}$$

where $\epsilon_t \sim N(0, 1)$. The red term is **random shock that scales with** $\sqrt{t}$.

The Square Root Law"Uncertainty accumulates as $\sqrt{t}$, not t"$\text{Var}(W_T) = T$"Interview: Why does vol scale with $\sqrt{t}$? Because variance is additive: independent shocks accumulate linearly in variance, so standard deviation accumulates as $\sqrt{t}$."

1.2 Why $\sqrt{t}$ Scaling? The Central Mystery

Random Walk Intuition:

$$W_T = \sum_{i=1}^{N} \Delta W_i, \quad \Delta W_i \sim N(0, \Delta t), \quad N = T/\Delta t$$

$$\text{Var}(W_T) = \sum_{i=1}^{N} \Delta t = N \Delta t = T$$

Core Formula

$$\text{SD}(W_T) = \sqrt{T}$$

"Variance is Additive, Volatility is Sub-Additive" **Memory:** 1-day variance = σ^2. 2-day variance = $2\sigma^2$. 2-day vol = $\sqrt{2}\sigma \approx 1.4\sigma$. **Not** 2σ.

Financial Interpretation: - **Day 1:** Order flow noise $\pm 1\%$ - **Day 2:** Independent noise $\pm 1\%$ - **Two-day move:** Not $\pm 2\%$, but $\pm 1.4\%$ on average (sum of squares)

The $\sqrt{t}$ Rule"Vol scales with square root of time because shocks are independent"$\text{Vol}(T) = \sigma\sqrt{T}$"This is why we use Δt in SDEs, not $(\Delta t)^2$. It's the fundamental scaling law of markets."

1.3 From Discrete to Continuous: The Limit

Discrete-Time Model:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t}\, Z_t, \quad Z_t \sim N(0, 1)$$

Module 9: Capstone—From Model to Price (End-to-End Flow)

"This Module Ties Everything Together. Master This, You Master Stochastic Calculus."

Step 1: Choose SDE Based on Economics

Example: Pricing a 5-year Bermudan swaption - Need: Positive rates, mean-reversion, some volatility clustering - Choice: 2-factor Hull-White short rate model

Core Formula

$$dr_t = \kappa(\theta_t - r_t)dt + \sigma dW_t$$

Why: Captures rate levels and some vol, fast enough for calibration.

Step 2: Apply Itô's Lemma to Derive Dynamics

For a function $V(r, t)$:

Core Formula

$$dV = V_t dt + V_r dr + \frac{1}{2} V_{rr} \sigma^2 dt$$

This prepares us for PDE or MC simulation.

Step 3: Delta-Hedge to Remove Randomness

For swaption, hedge with underlying swap:

Core Formula

$$\Pi = V - \Delta \cdot \mathbf{Swap}(r, t)$$

Choose $\Delta = \frac{\partial V}{\partial r}$ to cancel dW_t term.

Step 4: Derive Pricing PDE or Monte Carlo Scheme

- PDE Route: Solve backward from expiry:

Core Formula

$$V_t + \kappa(\theta - r)V_r + \frac{1}{2}\sigma^2 V_{rr} - rV = 0$$

- MC Route: Simulate forward:

Core Formula

$$r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma\sqrt{\Delta t} Z_t$$

Step 5: Apply Boundary Conditions

- Expiry: Swaption payoff = max(0, SwapRate(T) - K) - Bermudan: At each exercise date, $V_t = \max($Exercise Value, Con

Step 6: Compute Greeks Hedge Ratios

- Delta: $\Delta = \frac{\partial V}{\partial r}$ (DV01) - Vega: $\frac{\partial V}{\partial \sigma}$ - Gamma: $\frac{\partial^2 V}{\partial r^2}$

Step 7: Understand Model Limitations

- What it misses: Stochastic vol (smile), jumps (crisis), basis risk
- Check: Can rates go negative? (In HW, yes. In CIR, no)
- Validate: Backtest against historical exercise decisions

End-to-End Logic"Economics $\to$ SDE $\to$ Itô $\to$ Hedge $\to$ PDE/MC $\to$ Greeks $\to$ Validate"Each step is necessary, none optional"Interview: 'I can walk you through pricing a Bermudan swaption: choose HW model for rates, derive PDE via Itô, solve with finite difference, compute key-rate DV01 Greeks, and validate against market prices.'"

9.1 Common Pitfalls in End-to-End Pricing

Pitfall 1: Choosing Model Based on Fancy Math, Not Economics

Mistake: "Let's use Heston for rates because it's sophisticated."
Why Wrong: Heston designed for vol, not rates. Calibrates poorly to yield curve.
Right: Use Hull-White or LMM for rates. Use Heston for equity vol.

Pitfall 2: Forgetting to Check Boundary Conditions

Mistake: Solve PDE but forget $V(S=0,t)=0$ for call.
Why Wrong: Solution violates economics (can have negative price).
Right: Always list boundary conditions before solving. They define the contract.

Pitfall 3: Ignoring Hedging Error in Greeks

Mistake: Quote Greeks as if hedging is continuous.
Why Wrong: Real hedging error is large for high Gamma.
Right: Add reserve: Actual Delta $= \Delta_{\text{model}} \pm \text{HedgeError}(\Gamma, \Delta t)$

10.4 Pressure Questions (Test Composure & Deep Synthesis)

1. "I ran a Monte Carlo on my Heston model and got a price of \$5.12. My colleague ran the same model and got \$5.08. Both are 'correct'. Explain why." Model uncertainty, seeds, discretization. Answer: "Monte Carlo error. Standard error of 0.02 means prices differ by 1 standard error. Could be: (1) Different random seeds $\to$ different paths. (2) Different discretization (Euler vs Milstein). (3) Different variance reduction. (4) Different time steps. The 'correct' price is a distribution, not a point. I'd run both until standard error < 0.01 and see if they converge. If they don't, there's a bug in one implementation."

2. "Your delta hedge lost \$500k yesterday. The market moved 2%. Was it a good hedge or bad hedge?" Hedge quality vs outcome. Answer: "Can't judge by outcome alone. If model predicted 2% move $\to$ \$400k loss, then losing \$500k is within 1 std dev of hedging error (due to gamma, discrete rebalancing, vol mismatch). But if model predicted \$100k loss, then something's wrong: maybe gap, wrong vol assumption, or model breakdown. I'd check: (1) realized vs implied vol, (2) gamma exposure, (3) transaction costs. Good hedge can lose due to randomness; bad hedge loses systematically."

3. "Explain Itô's Lemma to me while you're stressed and haven't slept. No formulas, just raw intuition." Communication under pressure. Answer: "Okay, think of a function of a stock price. In normal calculus, you expand it like a Taylor series and ignore second-order terms because they're tiny. But stock prices are insane—they're super rough, like a drunkard's walk. So those 'tiny' second-order terms don't vanish—they add up to